

# Corner Subgrid Count

Time limit: 1.00 s

Memory limit: 512 MB

You are given an  $n \times n$  grid whose each square is either black or white. A subgrid is called *beautiful* if its height and width is at least two and all of its corners are black. How many beautiful subgrids are there within the given grid?

## Input

The first input line has an integer  $n$ : the size of the grid.

Then there are  $n$  lines describing the grid: 1 means that a square is black and 0 means it is white.

## Output

Print the number of beautiful subgrids.

## Constraints

- $1 \leq n \leq 3000$

## Example

| Input                                          | Output |
|------------------------------------------------|--------|
| 5<br>00010<br>11111<br>00110<br>11001<br>00010 | 4      |